

Corrections with Safe Reversion Across Estimation Tasks: What a Held-Out Check Can and Cannot Guarantee

Luciano Silva Alarco
Pontificia Universidad Católica del Perú
ORCID 0000-0002-3395-3129

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Abstract

Research line OMR001 proposed *correction operators* that revert to a *safe reference* when a held-out check fails, without showing differentiation from existing constructs. In the Gaussian location model we prove what it can and cannot guarantee. (i) If the source law is free of the target parameter, no correction improves the minimax risk, and for $d \leq 2$ every rule that ever corrects is worse than the reference somewhere. (ii) With a one-sided level- α check on m held-out observations, the excess over the reference is at most $\min\{\alpha\mathbb{E}[\Delta^+], \kappa(\alpha)(2\sigma/\sqrt{m})\mathbb{E}\|C - R\|\}$, with the best constant $\kappa(\alpha)$, and at most a cap uniform in the parameter, of order $m^{-1/2}$ for a fixed estimation sample; the correction is used and harmful with probability at most $\alpha\mathbb{P}(\Delta > 0)$. (iii) Reversion is penalised hold-out selection between two candidates, with a guarantee indifferent to the operator. (iv) Transporting the checked correction to the full-data mean removes the additive split cost from the bound, which becomes proportional to the correction, but not from the risk: even the oracle correction $C = \theta$ can lose to the split, or to the full-data mean for large m , and uniformly in C at least 4α times the split cost remains; cross-fitting inherits the bounds. In simulation every bound holds; the guaranteed construction is useful (predefined criterion) only for close tasks, because the split costs 25.0% of the full-data risk. Nothing is claimed about general accuracy or transfer guarantees.

1 Introduction

Suppose an analyst estimates the same kind of quantity, a mean or a regression coefficient, in a family of related tasks. Each task has its own *reference* estimator, computed from that task’s data alone and trusted because it is unbiased or minimax. A *correction operator* maps the reference estimate to a corrected one using what was learned on the other tasks: shrinkage toward a pooled centre, an additive bias adjustment, a linear map fitted on source tasks. Since the correction can hurt when the target task does not share the structure, research line OMR001 proposed to *revert* to the reference whenever a held-out check on the target does not favour the correction. Its public record states the objective (evaluate corrections on held-out tasks; conditions for transfer), the finding (no sufficient differentiation from existing model combinations and operator-based corrections) and the scope (no general accuracy or transfer guarantee).

The original definitions of the operators and of the “safe reference” are not available. This note therefore reconstructs a self-contained version of the construction (Section 2) and asks the question the line left open: *what, exactly, can a correction with safe reversion guarantee, and what can it not?* The answers are elementary but, we think, worth having in one place with proofs and with their constants.

Section 3 proves what *cannot* be guaranteed (any gain is a statement about average risk over tasks), and Section 4 what *can*: a do-no-harm bound of order $\sigma\mathbb{E}\|C - R\|/\sqrt{m}$ with the best constant for one-sided checks at a fixed level, a cap uniform in the parameter, of exact order $m^{-1/2}$ for a fixed estimation sample, a bound on harmful transfer events, and near-oracle selection. Section 5 asks whether the split can be undone by refitting after the check: with the correction transported to the full-data mean the additive split cost leaves the bound but not the risk; even the oracle correction can lose to the split, and in the worst case at least 4α times the split cost remains. Section 6 shows that the reverting estimator is penalised hold-out selection between two candidates, so that its safety is a property of selection and not of the operator, which explains why no differentiation should have been expected from the safety mechanism itself. Section 7 checks the bounds and reports, under a predefined criterion, where “useful transfer” occurs.

Every mathematical statement below carries a status tag in Table 5: proved here, classical, from the literature, computationally verified, or empirical only. Every number in the text is generated from the results of one script with a fixed seed (Appendix A).

2 Setting and definitions

Tasks and data. A task is a parameter $\theta \in \mathbb{R}^d$ together with observations whose law depends on θ . We work throughout in the Gaussian location model with known variance: an observation of task θ is $Y \sim N(\theta, \sigma^2 I_d)$.

The *target* task has parameter θ and provides $n = n_e + m$ independent observations; the first n_e form the *estimation sample*, with mean $\bar{X}_e \sim N(\theta, \sigma_e^2 I_d)$, $\sigma_e^2 = \sigma^2/n_e$, and the last m form the *held-out sample* $Y_1, \dots, Y_m$, with mean $\bar{Y} \sim N(\theta, \sigma^2 I_d/m)$. *Source* tasks $s = 1, \dots, S$ have parameters θ_s and produce data D whose law may or may not carry information about θ ; this is the whole question. We write $\mathcal{F}$ for the σ -field generated by the estimation sample and by D ; the held-out sample is independent of $\mathcal{F}$ given θ .

Loss and risk. For an estimate δ of θ the loss is $\ell(\delta) = \|\delta - \theta\|^2$ and the risk is $\mathcal{R}(\theta, \delta) = \mathbb{E}_\theta \ell(\delta)$. When θ is itself drawn from a distribution over tasks, the average of the risk is the Bayes risk.

Definition 2.1 (Reference, correction, reversion). (a) The *reference* estimator is the task-specific mean $R = \bar{X}_e$; it is unbiased and minimax for $\theta \in \mathbb{R}^d$ (Theorem 3.1(a)).

(b) A *correction operator* is an $\mathcal{F}$ -measurable map producing a corrected estimate $C = C(R; D) \in \mathbb{R}^d$. Three instances are used below: the affine shrinkage $C = \hat{\mu} + \lambda(R - \hat{\mu})$ toward a centre $\hat{\mu}$ estimated from the sources, with λ either estimated from the sources (the empirical Bayes rule of Efron and Morris [6, 7], Morris [15]), fixed in advance, or equal to zero (full pooling).

(c) The *held-out check* compares the held-out losses of the two candidates,

$$\hat{D} = \|C - \bar{Y}\|^2 - \|R - \bar{Y}\|^2,$$

and the *reverting estimator* with margin $c \geq 0$ ($\mathcal{F}$ -measurable) is

$$\hat{\theta}_{\text{rev}} = C \mathbf{1}\{\hat{D} \leq -c\} + R \mathbf{1}\{\hat{D} > -c\}.$$

The margin $c = z_{1-\alpha} s$ with $s = 2\sigma\|C - R\|/\sqrt{m}$ and $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$ makes the check a one-sided test at level α of “the correction does not help” (Lemma 4.1); $c = 0$ is plain hold-out selection.

We write $\Delta = \ell(C) - \ell(R)$ for the loss increase caused by the correction (negative when it helps), $\Delta^+ = \max(\Delta, 0)$, and $\pi = \mathbb{P}(\hat{D} \leq -c \mid \mathcal{F}, \theta)$ for the conditional probability that the check passes.

Shared structure. In the simulation the parameters of all tasks are exchangeable draws from $N(\mu, \tau^2 I_d)$ (a hierarchical or empirical Bayes model). The amount of shared structure is measured by $\text{snr} = \tau^2/\sigma_e^2$: at $\text{snr} = 0$ all tasks have the same parameter, at $\text{snr} \rightarrow \infty$ they share nothing. Departure from the structure is modelled by shifting the target’s centre by $\kappa\tau$ along one coordinate while the sources stay put.

3 What cannot be guaranteed

Theorem 3.1 (No free lunch for corrections). *Let $X \sim N(\theta, \sigma_e^2 I_d)$ with $\theta \in \mathbb{R}^d$ unknown and σ_e^2 known, and let D be auxiliary data, independent of X given θ , whose law Q does not depend on θ . Let $\delta = \delta(X, D)$ be any estimator; D may include auxiliary randomisation, so randomised rules are covered, and an estimator with infinite risk satisfies (a) and (c) trivially.*

(a) $\sup_\theta \mathcal{R}(\theta, \delta) \geq d\sigma_e^2 = \sup_\theta \mathcal{R}(\theta, X)$. [classical; proof included]

(b) If $\delta = X - b(D)$ for a measurable b with $\mathbb{E}\|b(D)\|^2 < \infty$, then $\mathcal{R}(\theta, \delta) = d\sigma_e^2 + \mathbb{E}\|b(D)\|^2$ for every θ . [proved here]

(c) If $d \leq 2$ and $\mathbb{P}_\theta(\delta \neq X) > 0$ for some (equivalently every) θ , then $\mathcal{R}(\theta', \delta) > d\sigma_e^2$ for some θ' . [literature (admissibility) + proved here]

(d) If $d \geq 3$ there exist δ with $\mathcal{R}(\theta, \delta) < d\sigma_e^2$ for every θ ; by (a) the supremum is still $d\sigma_e^2$. [literature]

Proof. (a) Fix $\tau > 0$ and put the prior $\theta \sim N(0, \tau^2 I_d)$, so that (θ, X, D) has joint law $\pi(d\theta)P_\theta(dx)Q(dD)$. Because Q does not depend on θ and D is independent of X given θ , the conditional law of θ given (X, D) equals its conditional law given X , namely $N(\frac{\tau^2}{\tau^2 + \sigma_e^2}X, \frac{\tau^2\sigma_e^2}{\tau^2 + \sigma_e^2}I_d)$. The Bayes risk of any $\delta(X, D)$ is therefore at least that of the posterior mean, $d\tau^2\sigma_e^2/(\tau^2 + \sigma_e^2)$. Since the supremum of the risk dominates its average, $\sup_\theta \mathcal{R}(\theta, \delta) \geq d\tau^2\sigma_e^2/(\tau^2 + \sigma_e^2)$ for every τ ; let $\tau \rightarrow \infty$. Equality for $\delta = X$ is direct. (b) $\mathbb{E}_\theta\|X - b - \theta\|^2 = \mathbb{E}\|X - \theta\|^2 + \mathbb{E}\|b\|^2 - 2\mathbb{E}\langle X - \theta, b \rangle$; the cross term is finite by Cauchy–Schwarz and vanishes because $X - \theta$ has mean zero and is independent of D . (c) Let $\tilde{\delta}(X) = \mathbb{E}[\delta(X, D) \mid X]$, which does not depend on θ because the conditional law of D given X is Q . Since $\tilde{\delta}$ is the conditional expectation of δ given X and $\tilde{\delta} - \theta$ is X -measurable, $\mathcal{R}(\theta, \delta) = \mathcal{R}(\theta, \tilde{\delta}) + \mathbb{E}_\theta\|\delta - \tilde{\delta}\|^2$ (Pythagoras; we may assume the left side finite). If $\mathbb{P}(\tilde{\delta} \neq X) > 0$, suppose for contradiction that $\mathcal{R}(\theta, \tilde{\delta}) \leq d\sigma_e^2$ for all θ ; then $\delta' = (\tilde{\delta} + X)/2$ satisfies $\|\delta' - \theta\|^2 = \frac{1}{2}\|\tilde{\delta} - \theta\|^2 + \frac{1}{2}\|X - \theta\|^2 - \frac{1}{4}\|\tilde{\delta} - X\|^2$, hence $\mathcal{R}(\theta, \delta') \leq d\sigma_e^2 - \frac{1}{4}\mathbb{E}_\theta\|\tilde{\delta} - X\|^2 < d\sigma_e^2$ for every θ (the Gaussian laws P_θ are mutually absolutely continuous, so the last expectation is positive for all θ once it is positive for one). This contradicts the admissibility of X for $d \leq 2$ [3, 17]; see Lehmann and Casella [12, Ch. 5]. If instead $\tilde{\delta} = X$ a.s., then $\mathcal{R}(\theta, \delta) = d\sigma_e^2 + \mathbb{E}_\theta\|\delta - X\|^2 > d\sigma_e^2$ for every θ . (d) is the James–Stein theorem [11, 17]; the risk of the James–Stein estimator tends to $d\sigma_e^2$ as $\|\theta\| \rightarrow \infty$. $\square$

Remark 3.2 (Where transfer can enter). The only hypothesis of Theorem 3.1(a) that a transfer setting can violate is that the law of the source data does not depend on θ . When θ and the source parameters are exchangeable draws from $N(\mu, \tau^2 I_d)$ with (μ, τ^2) fixed, the law of the source data still does not depend on the realised θ , so part (a) applies to the frequentist risk even in the hierarchical model; the sources are informative about the task distribution, which is what the Bayes risk over tasks rewards, and that is the natural criterion. For the affine correction $C = \mu + \lambda(X - \mu)$ with μ and $\lambda \in [0, 1)$ fixed,

$$\mathcal{R}(\theta, C) = \lambda^2 d\sigma_e^2 + (1 - \lambda)^2 \|\theta - \mu\|^2, \quad \mathbb{E}_{\theta \sim N(\mu, \tau^2 I)} \mathcal{R}(\theta, C) = \lambda^2 d\sigma_e^2 + (1 - \lambda)^2 d\tau^2, \quad (1)$$

minimised at $\lambda^* = \tau^2 / (\tau^2 + \sigma_e^2)$ with value $d\sigma_e^2 \lambda^* < d\sigma_e^2$. [classical] The available relative gain over the reference is $1 - \lambda^* = 1/(1 + \text{snr})$. The first formula in (1) is the negative-transfer risk: at a target far from the centre the correction is worse than the reference by $(1 - \lambda)^2 \|\theta - \mu\|^2 - (1 - \lambda^2) d\sigma_e^2$, unbounded in θ . Reversion is meant to cap this term. Section 4 gives two bounds: one of order $\mathbb{E}\|C - R\|/\sqrt{m}$, which for the shrinkage operator grows linearly with $\|\theta - \mu\|$ and so trades a quadratic harm for a linear one, and a cap uniform in θ (Corollary 4.3).

4 What can be guaranteed

Lemma 4.1 (The held-out loss difference). *With the notation of Definition 2.1, $\widehat{D} - \Delta = -2\langle C - R, \bar{Y} - \theta \rangle$. Hence $\mathbb{E}[\widehat{D} \mid \mathcal{F}, \theta] = \Delta$, and if the held-out observations are Gaussian then, conditionally on $(\mathcal{F}, \theta)$, $\widehat{D} \sim N(\Delta, s^2)$ with $s = 2\sigma\|C - R\|/\sqrt{m}$. In particular the rule “use C iff $\widehat{D} \leq -z_{1-\alpha}s$ ” rejects the hypothesis $\Delta \geq 0$ with conditional probability $\Phi(-(\Delta + z_{1-\alpha}s)/s) \leq \alpha$ whenever $\Delta \geq 0$.*

Proof. Expand $\|C - \bar{Y}\|^2 - \|R - \bar{Y}\|^2 = \|C\|^2 - \|R\|^2 - 2\langle C - R, \bar{Y} \rangle$ and $\Delta = \|C\|^2 - \|R\|^2 - 2\langle C - R, \theta \rangle$. Given $(\mathcal{F}, \theta)$, $\bar{Y} - \theta \sim N(0, \sigma^2 I_d/m)$ is independent of $C - R$, so $\langle C - R, \bar{Y} - \theta \rangle \sim N(0, \sigma^2 \|C - R\|^2/m)$. $\square$

Theorem 4.2 (Do no harm, up to a controlled amount). *Let R, C be $\mathcal{F}$ -measurable, let the held-out sample be independent of $\mathcal{F}$ given θ with mean θ , and let $\hat{\theta}_{\text{rev}}$ be the reverting estimator with margin $c \geq 0$. Write $s = 2\sigma\|C - R\|/\sqrt{m}$.*

(i) (Identity.) $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) = \mathbb{E}[\Delta \pi] \leq \mathbb{E}[\Delta^+ \pi]$, where $\pi = \mathbb{P}(\widehat{D} \leq -c \mid \mathcal{F}, \theta)$.

(ii) (Gaussian held-out, test at level $\alpha \in (0, \frac{1}{2}]$, $c = z_{1-\alpha}s$.) With $\kappa(\alpha) = \sup_{u \geq 0} u \Phi(-(z_{1-\alpha} + u)) \leq \varphi(z_{1-\alpha})$ (values in *results/tables.md*, T5; e.g. $\kappa(0.1) = 0.0188$),

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \leq \min \left\{ \alpha \mathbb{E}[\Delta^+], \quad \kappa(\alpha) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\| \right\},$$

and the probability of a harmful transfer event (C is used and $\Delta > 0$) satisfies $\mathbb{P}(\hat{\theta}_{\text{rev}} = C, \Delta > 0) \leq \alpha \mathbb{P}(\Delta > 0) \leq \alpha$; equivalently, given $\Delta > 0$, C is used with probability at most α .

(iii) (Bounded held-out.) If instead $\|Y_j - \theta\| \leq \rho$ a.s. and $c = 2\rho\|C - R\|\sqrt{2\log(1/\alpha)/m}$, then the same two statements hold with $\kappa(\alpha) 2\sigma/\sqrt{m}$ replaced by $2\rho/\sqrt{e}m$.

(iv) (Near-oracle selection, Gaussian held-out, $c = z_{1-\alpha}s$.)

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) \leq \mathbb{E}[\min\{\ell(R), \ell(C)\}] + (z_{1-\alpha} + \varphi(0)) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\|.$$

[proved here; standard arguments]

Proof. (i) $\ell(\hat{\theta}_{\text{rev}}) = \ell(R) + \Delta \mathbf{1}\{\widehat{D} \leq -c\}$. Take expectations conditionally on $(\mathcal{F}, \theta)$, on which Δ is measurable: $\mathbb{E}[\ell(\hat{\theta}_{\text{rev}}) - \ell(R) \mid \mathcal{F}, \theta] = \Delta \pi$. On $\{\Delta \leq 0\}$ this is ≤ 0 , whence the inequality.

(ii) By Lemma 4.1, on $\{\Delta > 0\}$ we have $\pi = \Phi(-(z + u))$ with $z = z_{1-\alpha} \geq 0$ and $u = \Delta/s \geq 0$ (if $s = 0$ then $C = R$, $\Delta = 0$ and there is nothing to prove). First bound: $\Phi(-(z + u)) \leq \Phi(-z) = \alpha$, so $\mathbb{E}[\Delta^+ \pi] \leq \alpha \mathbb{E}[\Delta^+]$. Second bound: $\Delta^+ \pi = s u \Phi(-(z + u)) \leq s \kappa(\alpha)$ by definition of κ ; take expectations. By the Mills ratio inequality $\Phi(-x) \leq \varphi(x)/x$ for $x > 0$ and since φ decreases on $[0, \infty)$,

$$u \Phi(-(z + u)) \leq \frac{u}{z + u} \varphi(z + u) \leq \varphi(z + u) \leq \varphi(z),$$

so $\kappa(\alpha) \leq \varphi(z)$. For the harmful event, $\mathbb{P}(\hat{\theta}_{\text{rev}} = C, \Delta > 0) = \mathbb{E}[\mathbf{1}\{\Delta > 0\} \pi] \leq \alpha \mathbb{P}(\Delta > 0)$.

(iii) Write $\widehat{D} - \Delta = \frac{1}{m} \sum_j W_j$ with $W_j = -2\langle C - R, Y_j - \theta \rangle$, independent given $(\mathcal{F}, \theta)$, mean zero and $|W_j| \leq B := 2\rho\|C - R\|$. Hoeffding’s inequality [9] gives $\mathbb{P}(\widehat{D} - \Delta \leq -t \mid \mathcal{F}, \theta) \leq \exp(-mt^2/(2B^2))$. On $\{\Delta \geq 0\}$, $\pi \leq \exp(-m(c + \Delta)^2/(2B^2)) \leq \exp(-mc^2/(2B^2)) = \alpha$ for the stated c , which yields the α -bound and the bound on the harmful event. Also $\Delta^+ \pi \leq \Delta^+ \exp(-m(\Delta^+)^2/(2B^2)) \leq \sup_{x \geq 0} x e^{-mx^2/(2B^2)} = B/\sqrt{em}$.

(iv) $\ell(\hat{\theta}_{\text{rev}}) - \min\{\ell(R), \ell(C)\} = |\Delta| \mathbf{1}\{\text{wrong choice}\}$, where the choice is wrong on $\{\Delta > 0, \hat{D} \leq -c\}$ and on $\{\Delta < 0, \hat{D} > -c\}$. Conditionally on $(\mathcal{F}, \theta)$, with $u = |\Delta|/s$: on $\{\Delta > 0\}$ the conditional expectation is $s u \Phi(-(z+u)) \leq s \varphi(z+u) \leq s \varphi(0)$ as in (ii); on $\{\Delta < 0\}$ it is $s u \Phi(z-u)$ by Lemma 4.1, which is at most $s z$ when $u \leq z$ and, writing $u = z + v$ with $v > 0$, at most $s(z + v \Phi(-v)) \leq s(z + \varphi(v)) \leq s(z + \varphi(0))$ otherwise. Both cases are bounded by $s(z + \varphi(0))$; take expectations. $\square$

Corollary 4.3 (A cap uniform in the parameter). *Under the hypotheses of Theorem 4.2(ii), with $R = \bar{X}_e$ and any $\mathcal{F}$ -measurable C , for every θ ,*

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \leq \frac{4\sigma}{\sqrt{m}} \kappa(\alpha) \mathbb{E}\|R - \theta\| + \frac{4\sigma^2}{m} \kappa_2(\alpha) \leq 4\sigma^2 \left(\kappa(\alpha) \sqrt{\frac{d}{n_e m}} + \frac{\kappa_2(\alpha)}{m} \right),$$

where $\kappa_2(\alpha) = \sup_{u \geq 0} u^2 \Phi(-(z_{1-\alpha} + u)) \leq \varphi(1) = \sup_{x \geq 0} x \varphi(x) = 0.2420$, so that $(\varphi(z_{1-\alpha}), \varphi(1))$ may replace (κ, κ_2) as a closed form. Theorem 4.2(ii) is sharper when $\mathbb{E}\|C - R\|$ is small; the cap takes over when it is large, where the bound of Theorem 4.2(ii) grows without limit although the harm does not. [proved here]

Proof. Write $a = \|C - R\|$, $e = R - \theta$ and $z = z_{1-\alpha}$, so that $\Delta = \|(C - R) + e\|^2 - \|e\|^2 = a^2 + 2\langle C - R, e \rangle \geq a(a - 2\|e\|)$; on $\{\Delta > 0\}$ put $u = \Delta/s$, so that $\Delta^+ \pi = s u \Phi(-(z+u))$. If $a \leq 2\|e\|$, then $s \leq 4\sigma\|e\|/\sqrt{m}$ and $\Delta^+ \pi \leq s \kappa \leq (4\sigma\|e\|/\sqrt{m})\kappa$. If $a > 2\|e\|$, then $u \geq w := \sqrt{m}(a - 2\|e\|)/(2\sigma) > 0$ and $s = (2\sigma/\sqrt{m})a = 4\sigma\|e\|/\sqrt{m} + 4\sigma^2 w/m$, so, because $w \leq u$,

$$\Delta^+ \pi = \frac{4\sigma\|e\|}{\sqrt{m}} u \Phi(-(z+u)) + \frac{4\sigma^2}{m} w u \Phi(-(z+u)) \leq \frac{4\sigma\|e\|}{\sqrt{m}} \kappa + \frac{4\sigma^2}{m} \kappa_2.$$

Take expectations in Theorem 4.2(i) and use $\mathbb{E}\|e\| \leq (\mathbb{E}\|e\|^2)^{1/2} = \sigma\sqrt{d/n_e}$. Finally, by the Mills ratio and $z \geq 0$, $u^2 \Phi(-(z+u)) \leq u \varphi(z+u) \leq (z+u) \varphi(z+u) \leq \varphi(1)$. Nothing is assumed about C beyond measurability, and the right side does not depend on θ . $\square$

Corollary 4.4 (Price of the split). *Let $R_n = \bar{X}_n$ be the reference computed from all $n = n_e + m$ target observations, with risk $d\sigma^2/n$. Then*

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R_n) = d\sigma^2 \left(\frac{1}{n_e} - \frac{1}{n} \right) + (\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R)),$$

so the total cost of the safety mechanism relative to “do nothing with all the data” is the split cost $d\sigma^2 m/(nn_e)$, a fraction m/n_e of the full-data risk $d\sigma^2/n$ (equivalently m/n of the risk of R), plus the controlled term of Theorem 4.2. [proved here (arithmetic)]

Proposition 4.5 (Sharpness). *Fix $\alpha \in (0, \frac{1}{2}]$ and the check of Theorem 4.2(ii), and let u^* attain $\kappa(\alpha)$.*

- (a) (Best constant.) *For every $\varepsilon > 0$ there are a Gaussian target model and an $\mathcal{F}$ -measurable pair (R, C) with $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \geq (\kappa(\alpha) - \varepsilon) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\|$. Thus $\kappa(\alpha)$ is the best constant in Theorem 4.2(ii); the closed form $\varphi(z_{1-\alpha})$ loses the factor $\varphi(z_{1-\alpha})/\kappa(\alpha)$, from 2.3 to 20 on the grid of Section 7.*
- (b) (Rate for a fixed estimation sample.) *Let $R = \bar{X}_e$ and $c_d = \mathbb{E}\|Z\|$ for $Z \sim N(0, I_d)$, so that $\mathbb{E}\|R - \theta\| = \sigma c_d/\sqrt{n_e}$ and $c_d \leq \sqrt{d}$. For all d , n_e and m , the supremum over θ and over $\mathcal{F}$ -measurable C of $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R)$ lies between*

$$4\sigma^2 \left(\frac{\kappa(\alpha) c_d}{\sqrt{n_e m}} + \frac{u^* \kappa(\alpha)}{m} \right) \quad \text{and} \quad 4\sigma^2 \left(\frac{\kappa(\alpha) c_d}{\sqrt{n_e m}} + \frac{\kappa_2(\alpha)}{m} \right).$$

For a fixed estimation sample the worst-case harm is therefore of exact order $m^{-1/2}$, and the first term of Corollary 4.3 is attained; as $n_e \rightarrow \infty$ it is of order $1/m$.

- (c) (Exact worst case.) *With $\xi = \sqrt{m} \|R - \theta\|/\sigma \sim \sqrt{m/n_e} \chi_d$ and $M_0(\alpha; \xi) = 4 \sup_{u \geq 0} (u^2 + \xi u) \Phi(-(z_{1-\alpha} + u))$, the supremum in (b) equals $(\sigma^2/m) \mathbb{E}[M_0(\alpha; \xi)]$, at every θ ; by Corollary 4.4 the worst case of $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(\bar{X}_n)$ is $d\sigma^2 m/(nn_e) + (\sigma^2/m) \mathbb{E}[M_0(\alpha; \xi)]$. In the design of Section 7 at $\alpha = 0.1$ the former is 0.0100, inside the interval $[0.0096, 0.0106]$ of (b).*

Both parts concern this class of checks: a larger margin (in the limit, never using C) lowers the worst-case harm and pays through the regret term of Theorem 4.2(iv). The worst case is a correction small relative to the noise of the check ($\Delta/s = u^ \leq 0.75$). [proved here; constants and $\mathbb{E}[M_0]$ computed numerically; (b) also checked by Monte Carlo, Appendix A; (c) since v0.5, after the third internal review]*

Proof. (a) Take $R = \bar{X}_e$ and $C = R + v$ with a fixed vector $v \neq 0$ (an additive correction fixed in advance, as in Theorem 3.1(b)). Then $\|C - R\| = \|v\|$, $s = 2\sigma\|v\|/\sqrt{m}$ is deterministic and $\Delta = \|v\|^2 + 2\langle v, R - \theta \rangle$. By Theorem 4.2(i) the excess risk equals $\mathbb{E}[g(\Delta)]$ with $g(x) = x \Phi(-(x + zs)/s)$, a continuous function with $|g(x)| \leq |x|$. As $n_e \rightarrow \infty$, $\Delta \rightarrow \|v\|^2$ in L^2 , so $\mathbb{E}[g(\Delta)] \rightarrow g(\|v\|^2) = s u \Phi(-(z + u))$ with $u = \|v\|^2/s = \|v\|\sqrt{m}/(2\sigma)$, which can be given any prescribed value $u \geq 0$ by the choice of $\|v\|$. Choosing $u = u^*$ and n_e large gives the claim. (b) The upper bound is Corollary 4.3; the lower bound is the construction of (c) with $u \equiv u^*$, i.e. $C = \theta_0 - (1+t)(R - \theta_0)$ with $t = 2\sigma u^*/(\sqrt{m}\|R - \theta_0\|)$, whose excess at θ_0 is $\mathbb{E}[g(u^*, \xi)]\sigma^2/m = \kappa(\alpha)(4\sigma\mathbb{E}\|e\|/\sqrt{m} + 4\sigma^2 u^*/m)$ (also directly: (b) follows from (c) and $(u^2 + \xi u)\Phi(-(z + u)) \leq \kappa_2 + \xi\kappa$). (c) Write $e = R - \theta$, $\omega = \sqrt{m}\|C - R\|/\sigma$, $\delta = m\Delta/\sigma^2$ and, for $\omega > 0$, $u = \delta/(2\omega) = \Delta/s$. Then $\Delta\pi = (\sigma^2/m)2\omega u \Phi(-(z + u))$, and $|\Delta - \|C - R\|^2| = 2|\langle C - R, e \rangle| \leq 2\|C - R\|\|e\|$ reads $|2u - \omega| \leq 2\xi$. If $u \leq 0$ the conditional excess is ≤ 0 ; if $u > 0$ it is at most $(\sigma^2/m)g(u, \xi)$ with $g(u, \xi) = 4(u^2 + \xi u)\Phi(-(z + u)) \geq 0$, because $\omega \leq 2(u + \xi)$; so it is at most $(\sigma^2/m)M_0(\alpha; \xi)$; take expectations. Conversely, fix θ_0 and let $C = \theta_0 - (1 + 2u(\xi)/\xi)(R - \theta_0)$ with $u(\cdot) \geq 0$ measurable, an $\mathcal{F}$ -measurable function of R . At θ_0 , $\omega = 2(\xi + u(\xi))$ and $\Delta/s = u(\xi)$, so the conditional excess is exactly $(\sigma^2/m)g(u(\xi), \xi)$. Since $|g(u, \xi) - g(u, \xi')| \leq 4\kappa(\alpha)|\xi - \xi'|$, M_0 is 4κ -Lipschitz; taking u constant on each cell $[jh, (j+1)h)$, equal to a maximiser of $g(\cdot, jh)$ (it exists: $g(0, \xi) = 0$ and $g \rightarrow 0$ as $u \rightarrow \infty$), gives $g(u(\xi), \xi) \geq M_0(\alpha; \xi) - 8\kappa h$; let $h \rightarrow 0$. $\square$

Remark 4.6 (Detectability, not estimability; variants of the check). When the correction helps ($\Delta < 0$), Lemma 4.1 gives the conditional probability of using it, $\Phi(-z_{1-\alpha} + |\Delta|/s)$ with $|\Delta|/s = |\Delta|\sqrt{m}/(2\sigma\|C - R\|)$. In the shift example of Proposition 4.5 with a beneficial v that cancels the error of R exactly ($R - \theta = -v$, so $\Delta = -\|v\|^2$) this ratio is $\|v\|\sqrt{m}/(2\sigma)$ whatever the dimension: the check is a one-dimensional test along the direction $C - R$ supplied by the sources, so a gain can be *verified* on the target with a held-out sample that would be far too small to *learn* the correction there. Gains too small to be detected at level α are forfeited, up to the regret term of Theorem 4.2(iv). Variants: (a) if σ is unknown, the check can be the one-sided t -test at level α on the m projections $\langle C - R, Y_j \rangle$; under Gaussian held-out observations it is exact at $\Delta = 0$ with power monotone in the non-centrality [13], so the α -form of Theorem 4.2(ii) and the harmful-event bound hold verbatim. (b) Nothing uses that C is a function of R : all statements hold for any two $\mathcal{F}$ -measurable candidates.

5 Refitting after the check, and cross-fitting

Corollary 4.4 charges the split cost $d\sigma^2 m/(nn_e)$ to every correction, even $C = R$. Refitting on all n observations after the check reuses the held-out sample, so Theorem 4.2 does not apply; but the check sees that sample only through $\langle C - R, \bar{Y} - \theta \rangle$ (Lemma 4.1), and this one-dimensional dependence can be computed exactly. Write $\eta = m/n$, $w = C - R$, $e = R - \theta$, $s = 2\sigma\|w\|/\sqrt{m}$, $z = z_{1-\alpha}$, $u = \Delta/s$, $\pi = \Phi(-(z + u))$, $\xi = \sqrt{m}\|e\|/\sigma$, and $J = \mathbf{1}\{\hat{D} \leq -zs\}$ for the level- α check of Definition 2.1(c). The *refit with transported correction* adds the correction vector proposed on the estimation sample to the full-data mean:

$$\hat{\theta}_{\text{rf}} = \bar{X}_n + J(C - R).$$

For an additive correction $C = R + v$ it applies v to $\bar{X}_n$ itself. It is *not* the “refit” variant of Section 7, which re-applies the operator to $\bar{X}_n$ (for the EB operator with $\hat{\lambda}_n = \hat{\tau}^2/(\hat{\tau}^2 + \sigma^2/n)$ instead of $\hat{\lambda}$); that variant remains without a guarantee. Define, for $x, \xi \geq 0$, $\Psi_\eta(\alpha; x) = \sup_{u \in \mathbb{R}} \{(u + x)\Phi(-(z + u)) + \eta\varphi(z + u)\}$,

$$M_\eta(\alpha; \xi) = \sup_{\omega > 0, |\delta - \omega^2| \leq 2\xi\omega} f_\eta(\omega, \delta), \quad f_\eta(\omega, \delta) = ((1 - \eta)\delta + \eta\omega^2)\Phi(-(z + \frac{\delta}{2\omega})) + 2\eta\omega\varphi(z + \frac{\delta}{2\omega}),$$

and $\bar{\kappa}_\eta(\alpha) = \max\{\kappa(\alpha) + \eta\varphi(z), \eta\varphi(0)\}$, $\kappa_\varphi(\alpha) = \sup_{u \geq 0} u\varphi(z + u) \leq \varphi(1)$, $c_d = \mathbb{E}\|Z\|$ and $\mu_{3,d} = \mathbb{E}\|Z\|^3$ for $Z \sim N(0, I_d)$. By the proof of (iv), the supremum defining M_η may be restricted to $\delta = \omega^2 \pm 2\xi\omega$ (worst corrections collinear with $R - \theta$), so it is one-dimensional, and M_0 is the function of Proposition 4.5(c).

Theorem 5.1 (Refit with transported correction). *Let $R = \bar{X}_e$, let C be $\mathcal{F}$ -measurable, let the held-out observations be Gaussian and $\alpha \in (0, \frac{1}{2}]$. Then, for every θ :*

$$(i) \text{ (Identity.) } \mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) = \mathbb{E}\left[\left((1 - \eta)\Delta + \eta\|w\|^2\right)\pi + \eta s \varphi(z + u)\right] = \mathbb{E}[\Delta\pi] + \eta \mathbb{E}[s \varphi(z + u) - 2\langle w, e \rangle \pi].$$

(ii) (Bound proportional to the correction.)

$$\mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) \leq \mathbb{E}[s \Psi_\eta(\alpha; \eta\xi)] \leq \Psi_\eta(\alpha; 0) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\| + 2\eta \mathbb{E}[\|C - R\| \|R - \theta\|],$$

$$\text{and } \Psi_\eta(\alpha; x) \leq \bar{\kappa}_\eta(\alpha) + x \Phi(x - z).$$

(iii) (Cap uniform in θ and C .) $\mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) \leq (\sigma^2/m) \mathbb{E}[M_\eta(\alpha; \xi)]$, where $M_\eta(\alpha; \xi) \leq 4\xi\bar{\kappa}_\eta + 4\eta\xi^2\Phi(\eta\xi - z) + 4\kappa_2 + 4\eta(\xi\kappa + \kappa_\varphi)$. Hence

$$\mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) \leq \frac{4\sigma^2(\bar{\kappa}_\eta + \eta\kappa) c_d}{\sqrt{n_e m}} + 4\alpha \frac{d\sigma^2 m}{n n_e} + \frac{4\eta^2 \varphi(0) \mu_{3,d} \sigma^2 \sqrt{m}}{n_e^{3/2}} + \frac{4\sigma^2(\kappa_2 + \eta\kappa_\varphi)}{m}.$$

	$m = 3$		$m = 6$		$m = 12$		$m = 24$	
α	refit	split	refit	split	refit	split	refit	split
<i>worst case over θ and $\mathcal{F}$-measurable C</i>								
0.5	<i>0.3479</i>	0.3210	<i>0.2225</i>	0.1904	<i>0.1720</i>	0.1302	<i>0.1994</i>	0.1285
0.2	<i>0.0808</i>	0.0737	<i>0.0567</i>	0.0500	<i>0.0516</i>	0.0462	<i>0.0792</i>	0.0731
0.1	<i>0.0322</i>	0.0308	0.0241	0.0250	0.0244	0.0308	0.0450	0.0626
0.05	0.0138	0.0152	0.0109	0.0158	0.0123	0.0250	0.0269	0.0585
0.01	0.0022	0.0059	0.0020	0.0102	0.0028	0.0215	0.0090	0.0560
split cost	0.0044		0.0093		0.0208		0.0556	
<i>oracle correction $C = \theta$</i>								
0.5	<i>-0.0423</i>	-0.0495	<i>-0.0405</i>	-0.0522	<i>-0.0342</i>	-0.0558	<i>-0.0088</i>	-0.0614
0.2	<i>-0.0176</i>	-0.0212	<i>-0.0167</i>	-0.0222	<i>-0.0128</i>	-0.0230	<i>+0.0061</i> ^H	-0.0245
0.1	<i>-0.0089</i>	-0.0098	<i>-0.0083</i>	-0.0090	<i>-0.0054</i>	-0.0066	<i>+0.0098</i> ^H	-0.0012
0.05	-0.0045	-0.0034	-0.0040	-0.0012	-0.0019	+0.0041	+0.0101 ^H	+0.0166
0.01	-0.0009	+0.0025	-0.0007	+0.0065	+0.0002 ^H	+0.0158	+0.0063 ^H	+0.0405

Table 1: Exact excess risk over the full-data reference $\bar{X}_n$ of the transported refit and of the split estimator ($d = 5$, $n = 60$, $\sigma = 1$, $n_e = n - m$; quadrature, no Monte Carlo). Top: worst case over θ and $\mathcal{F}$ -measurable C , $(\sigma^2/m)\mathbb{E}[M_\eta(\alpha; \xi)]$ for the refit (Theorem 5.1(iii)–(iv)) and $d\sigma^2 m/(nn_e) + (\sigma^2/m)\mathbb{E}[M_0(\alpha; \xi)]$ for the split (Proposition 4.5(c)). Bottom: the oracle correction (Remark 5.3(e)); negative values are gains. Bold: refit better than split; italic: refit worse; ^H: refit harmful relative to $\bar{X}_n$.

(iv) (Attainment.) For every $d \geq 1$, every θ_0 and every $\varepsilon > 0$ there is an $\mathcal{F}$ -measurable $C = R + \lambda(R - \theta_0)$, with λ a function of $\|R - \theta_0\|$ (it inflates or shrinks the error of R , as in Proposition 4.5(c)), such that at $\theta = \theta_0$, $\mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) \geq (\sigma^2/m)\mathbb{E}[M_\eta(\alpha; \xi)] - \varepsilon$. Also, $C = 2\theta_0 - R$ gives at $\theta = \theta_0$ exactly $4\alpha d\sigma^2 m/(nn_e) + 4\eta\varphi(z)\sigma^2 c_d/\sqrt{n_e m}$; hence no bound uniform in C can be smaller than 4α times the split cost of Corollary 4.4, the multiple that appears in (iii).

[proved here (Appendix B); checked by an independent internal referee (round 3); Ψ_η , M_η computed numerically; checked by Monte Carlo]

Corollary 5.2 (Cross-fitted reversion). Let $n = Km$ and split the target sample into folds $I_1, \dots, I_K$ of size m . Let R_k be the mean outside I_k (so $n_e = n - m$), let C_k be measurable with respect to the sources and the data outside I_k , and let J_k be the level- α check of (R_k, C_k) on I_k . Then

$$\hat{\theta}_{\text{cf}} = \frac{1}{K} \sum_{k=1}^K (J_k C_k + (1 - J_k) R_k) = \bar{X}_n + \frac{1}{K} \sum_{k=1}^K J_k (C_k - R_k),$$

so $\hat{\theta}_{\text{cf}} = \bar{X}_n$ whenever no check passes, and $\mathcal{R}(\hat{\theta}_{\text{cf}}) - \mathcal{R}(\bar{X}_n)$ is at most the average over k of the bounds of Theorem 5.1(ii)–(iii) with $\eta = 1/K$; in particular it is at most $(\sigma^2/m)\mathbb{E}[M_{1/K}(\alpha; \xi)]$, uniformly in θ and in the C_k . [proved here (Jensen); checked by an independent internal referee (round 3)]

Remark 5.3 (What happens to the split cost). (a) *The bound has no split-cost term:* the bound of Theorem 5.1(ii) is the expectation of a quantity proportional to $\|C - R\|$ and is zero for $C = R$, whereas the split estimator pays $d\sigma^2 m/(nn_e)$ even then (Corollary 4.4). The price is a larger slope, $\Psi_\eta(\alpha; 0)$ against $\kappa(\alpha)$ (0.0405 against 0.0188 at $\alpha = 0.1$, $\eta = 0.2$), plus the term $2\eta\mathbb{E}[\|C - R\| \|R - \theta\|]$. This compares bounds, not risks: for a given correction the refit can be worse than the split, see (e). (b) *Uniformly in θ and C the split cost is only reduced,* and in the worst case at least the fraction 4α of it remains (Theorem 5.1(iv)). Table 1 compares the exact worst cases of $\hat{\theta}_{\text{rf}}$ and of the split estimator in the design of Section 7: the refit is better for $\alpha = 0.1$ ($m \in \{6, 12, 24\}$), $\alpha = 0.05$ (every m), $\alpha = 0.01$ (every m) and worse for $\alpha = 0.5$ (every m), $\alpha = 0.2$ (every m), $\alpha = 0.1$ ($m \in \{3\}$). With plain hold-out ($\alpha = 0.5$, $4\alpha = 2$) it is worse in this design and, numerically, in all 36 other designs we tried ($d \in \{1, 2, 5, 50\}$, $n_e \in \{2, 10, 1000\}$, $m \in \{1, 3, 30\}$; ratio 1.001–3.28); this is not proved. The closed form in (iii) is 1.08–13.6 times the exact cap on the grid; it does not vanish as $\alpha \rightarrow 0$ and is informative only for moderate α and $m \ll n_e$; for small α the exact cap should be used. (c) The route “use C if the check passes, otherwise $\bar{X}_n$ ” does not remove the split cost even for a vanishing correction $C = R - t(R - \theta_0)$, $t \rightarrow 0$: the check detects that C beats R , not that it beats $\bar{X}_n$, and the exact limit of the excess is 1.06 times the split cost at $m = 12$, $\alpha = 0.1$ (0.32–1.99 on the grid; derivation in `theory/refit_derivation.md`). (d) For cross-fitting the reflection of (iv) cancels in the average ($\frac{1}{K} \sum_k (R_k - \theta) = \bar{X}_n - \theta$), so the lower bound does not carry over; but cross-fitting is not free of the split cost either (up to 9.4 times it on the adversarial grid of Section 7), and the Jensen step is loose. (e) *Not better than the split for every correction.* For the oracle correction $C = \theta$ ($w = -e$, $\Delta = -\|e\|^2$, $u = -\xi/2$), identity (i) gives exactly

$$\mathcal{R}(\hat{\theta}_{\text{rf}}) - \mathcal{R}(\bar{X}_n) = \frac{\sigma^2}{m} \mathbb{E}[-(1 - 2\eta)\xi^2 \Phi(\xi/2 - z) + 2\eta\xi\varphi(z - \xi/2)],$$

against $d\sigma^2 m/(nn_e) - (\sigma^2/m)\mathbb{E}[\xi^2\Phi(\xi/2 - z)]$ for the split estimator (Table 1, bottom). At $\alpha = 0.1$ and $m = 12$ the split is better (-0.0066 against -0.0054); at $m = 24$ the transported oracle is harmful ($+0.0098$, 11.8% of the risk of $\bar{X}_n$) while the split gains (-0.0012). On the grid the split beats the refit with the oracle for $\alpha = 0.5$ (every m), $\alpha = 0.2$ (every m), $\alpha = 0.1$ (every m), and the refit is harmful for $\alpha = 0.2$ ($m \in \{24\}$), $\alpha = 0.1$ ($m \in \{24\}$), $\alpha = 0.05$ ($m \in \{24\}$), $\alpha = 0.01$ ($m \in \{12, 24\}$). Two effects combine: the check passes preferentially when the held-out noise points where it hurts $\bar{X}_n + w$ (the term $\eta\mathbb{E}[s\varphi(z + u)]$ of (i)), and $w = C - R$ corrects the error of R , not that of $\bar{X}_n$. If $m \geq n_e$ ($\eta \geq \frac{1}{2}$) both terms are non-negative, so the transported oracle is never better than $\bar{X}_n$.

6 Relation to known constructs, and the line’s conclusion

Proposition 6.1 (The reverting estimator is penalised hold-out selection). *Let $\hat{L}_m(\delta) = \frac{1}{m} \sum_{j=1}^m \|\delta - Y_j\|^2$ be the held-out empirical loss. Then $\hat{L}_m(C) - \hat{L}_m(R) = \hat{D}$, and*

$$\hat{\theta}_{\text{rev}} = \arg \min_{\delta \in \{R, C\}} \{ \hat{L}_m(\delta) + c \mathbf{1}\{\delta = C\} \}$$

(ties, a null event, resolved toward C as in Definition 2.1(c)). With $c = 0$ it is plain hold-out selection between two candidates. [proved here (one line)]

Proof. $\hat{L}_m(\delta) = \|\delta - \bar{Y}\|^2 + \frac{1}{m} \sum_j \|Y_j - \bar{Y}\|^2$, and the second term does not depend on δ . $\square$

Three consequences follow. First, the guarantee of Theorem 4.2 is the two-candidate case of the standard analysis of hold-out selection among finitely many candidates fixed before the held-out sample is seen [4, §8.4; see also Ch. 22], see also Arlot and Celisse [1]; with K candidates it costs an extra $\sqrt{\log K}$, and Theorem 4.2 only sharpens the Gaussian two-candidate constants and makes the one-sided reading explicit. Second, the correction operator enters the guarantee only through $\mathbb{E}\|C - R\|$, the size of the proposed change, and through Δ , its actual effect; the guarantee is indifferent to how the operator was obtained. Third, whether the reverting estimator *gains* anything is a question about the shared structure (Remark 3.2) and about whether that gain is detectable with m held-out points (Remark 4.6). Table 2 places the construction next to the classical ones it resembles.

construct	reverts to	trigger	guarantee
Positive-part James–Stein [2]	the shrinkage target	$\ X - \mu\ ^2 < (d - 2)\sigma_e^2$	dominates James–Stein; minimax, $d \geq 3$
Limited-translation rules [5]	the reference, coordinate-wise	component moved further than a cap	bounds the risk increase of each component
SURE selection [18]	either candidate	sign of an unbiased risk-difference estimate; no split	unbiased estimate; no finite-sample selection bound used here
Source detection in transfer learning [14]	target-only estimator	data-splitting aggregation on the target	rates under informative sources; protection against non-informative ones
Reverting estimator (this note)	the reference R	one-sided test at level α on $\hat{D}$	Theorem 4.2, Corollary 4.3: harm $\leq \alpha\mathbb{E}\Delta^+$, $\leq \kappa(\alpha) 2\sigma\mathbb{E}\ C - R\ /\sqrt{m}$ and $\leq$ a cap uniform in θ ; harmful event $\leq \alpha\mathbb{P}(\Delta > 0)$

Table 2: The reverting estimator next to constructs it coincides with or resembles. The first two revert in a different direction or by a different mechanism; the third is the same construct with a different check, and plain hold-out selection [1, 4] is the case $c = 0$ (Proposition 6.1); “negative transfer” is the name given to $\Delta > 0$ in the transfer-learning literature [16].

What this says about the line’s conclusion. Proposition 6.1 and Theorem 4.2 explain the line’s public finding (no differentiation from existing constructs): none should be expected *from the safety mechanism itself*: safe reversion *is* held-out selection between two candidates, its only guarantee is a selection guarantee, and that guarantee is available for any pair of candidates. Differentiation, if any, must come from the operator family, which this note does not explore beyond three affine instances, all empirical Bayes rules in the sense of Efron and Morris [6]; whether the line should be closed or continued along the open problems of Section 8 is a decision this note does not take. What it adds is the explicit guarantee with its best constant among fixed-level checks, the uniform cap with its exact worst case, and the price of the split (Sections 5 and 7).

7 Simulation

Protocol. $d = 5$, $\sigma = 1$, $S = 30$ source tasks with $n_s = 60$ observations each, a target task with $n = 60$ observations split into $n_e = 48$ for estimation and $m = 12$ for the check (a split cost of 25.0% of the full-data risk, 20.0% of the risk of R ; Corollary 4.4); $m \in \{3, 6, 12, 24\}$ is also swept, with τ^2 held at its $m = 12$ value ($\tau^2 = 0.0208 = \sigma^2/48$) so that only the split moves. Sources and target are exchangeable draws from $N(\mu, \tau^2 I_d)$

snr	λ^*	$\bar{X}_n$	always	reverting, with guarantee				without guarantee	
			all data	$\alpha=0.5$	$\alpha=0.2$	$\alpha=0.1$	$\alpha=0.01$	refit 0.1	SURE
0	0.00	0.0833	0.0030 ^U	0.0295 ^U	0.0610 ^U	0.0769 ^u	0.0996	0.0702 ^U	0.0176 ^U
0.25	0.20	0.0833	0.0233 ^U	0.0453 ^U	0.0697 ^U	0.0825	0.0998	0.0743 ^U	0.0365 ^U
0.5	0.33	0.0831	0.0347 ^U	0.0542 ^U	0.0745 ^U	0.0853	0.1002	0.0759 ^U	0.0449 ^U
1	0.50	0.0841	0.0485 ^U	0.0660 ^U	0.0815	0.0897	0.1015	0.0789 ^u	0.0562 ^U
2	0.67	0.0832	0.0605 ^U	0.0773 ^u	0.0877	0.0933	0.1015	0.0801	0.0653 ^U
4	0.80	0.0833	0.0699 ^U	0.0868	0.0935	0.0973	0.1033	0.0813	0.0728 ^U
8	0.89	0.0833	0.0761 ^U	0.0933	0.0973	0.0997	0.1034	0.0824	0.0776 ^u
16	0.94	0.0833	0.0797	0.0976	0.0998	0.1014	0.1037	0.0828	0.0804

Table 3: Structure sweep, EB operator, target drawn from the hierarchy, $m = 12$. Risks (mean squared error over $d = 5$ coordinates; Monte Carlo standard errors are below 0.0005). ^U marks configurations meeting the predefined criterion U (gain $\geq 5.0\%$ over the full-data reference with 95% confidence); ^u: borderline (lower end within 2 points of the threshold). The reference on the estimation sample has risk ≈ 0.1042 against 0.0833 for $\bar{X}_n$.

κ	$\bar{X}_n$	always	reverting, with guarantee				without guarantee	
		all data	$\alpha=0.5$	$\alpha=0.2$	$\alpha=0.1$	$\alpha=0.01$	refit 0.1	SURE
0	0.0837	0.0481	0.0655	0.0814	0.0895 ^H	0.1012 ^H	0.0785	0.0557
2	0.0842	0.0652	0.0792	0.0891 ^H	0.0947 ^H	0.1025 ^H	0.0822	0.0747
4	0.0831	0.1174 ^H	0.1074 ^H	0.1021 ^H	0.1017 ^H	0.1031 ^H	0.0859 ^H	0.0939 ^H
6	0.0837	0.2024 ^H	0.1340 ^H	0.1112 ^H	0.1070 ^H	0.1048 ^H	0.0888 ^H	0.0869 ^H
8	0.0834	0.3225 ^H	0.1528 ^H	0.1160 ^H	0.1084 ^H	0.1043 ^H	0.0890 ^H	0.0835 ^H
12	0.0838	0.6665 ^H	0.1621 ^H	0.1153 ^H	0.1085 ^H	0.1050 ^H	0.0877 ^H	0.0838
16	0.0835	1.1512 ^H	0.1454 ^H	0.1098 ^H	0.1054 ^H	0.1042 ^H	0.0847 ^H	0.0835

Table 4: Departure sweep, EB operator, $\text{snr} = 1$, target centre shifted by $\kappa\tau$, $m = 12$. Risks as in Table 3; Monte Carlo standard errors are below 0.0028. ^H: harmful relative to $\bar{X}_n$ (upper end of the 95% interval of the gain below zero).

with $\text{snr} = \tau^2/\sigma_e^2 \in \{0, 0.25, 0.5, 1, 2, 4, 8, 16\}$ (structure sweep), and, at $\text{snr} = 1$, the target centre is shifted by $\kappa\tau$ along one coordinate with $\kappa \in \{0, 2, 4, 6, 8, 12, 16\}$ (departure sweep). The operators are the empirical Bayes shrinkage $C = \hat{\mu} + \hat{\lambda}(R - \hat{\mu})$ with $\hat{\mu}$ the mean of the source means and $\hat{\lambda} = \hat{\tau}^2/(\hat{\tau}^2 + \sigma_e^2)$, $\hat{\tau}^2 = (\hat{v} - \sigma^2/n_s)^+$, $\hat{v}$ the pooled variance of the source means; full pooling ($\lambda = 0$); and a fixed $\lambda = \frac{1}{2}$. The check uses $\alpha \in \{0.5, 0.2, 0.1, 0.05, 0.01\}$. Each configuration uses 20 000 replicates with common random numbers across α and operators, seed 20260930. Two variants *without* a guarantee proved here are also run: *refit*, which applies the same check but then builds the final estimate from all n observations, so that the independence hypothesis of Theorem 4.2 fails, and *SURE reversion*, which uses no split and applies C to $\bar{X}_n$ iff Stein’s unbiased estimate [18] of $\mathcal{R}(C) - \mathcal{R}(\bar{X}_n)$ is negative. Since v0.4 the *transported refit* $\bar{X}_n + J(C - R)$ of Section 5, which has the guarantee of Theorem 5.1, is computed on the same replicates (it needs no extra draws), and the cross-fitted estimator of Corollary 5.2 with $K = n/m$ folds is run on fresh draws (seed 20260930 + 7) by `theory/check_refit.py`.

Predefined criteria. (U) *Useful transfer* at a configuration: the lower end of the paired 95% confidence interval of the relative risk gain with respect to the *full-data* reference $\bar{X}_n$ is at least 5.0%; the split is charged to the method. (S) *Safety*: the Monte Carlo excess risk of the reverting estimator over R never exceeds the identity bound, the α -bound or the closed form $\varphi(z)\mathbb{E}[s]$ of Theorem 4.2(i)–(ii) by more than two Monte Carlo standard errors, and the harmful-event frequency never exceeds $\alpha\hat{\mathbb{P}}(\Delta > 0)$ by more than two binomial standard errors. (H) The frequency of harmful transfer events, $\hat{\theta}_{\text{rev}} = C$ and $\Delta > 0$ (the correction is used and its realised loss exceeds that of the reference), is reported for every configuration. Criterion U was fixed in writing before the runs; Appendix A states what was chosen or changed later.

Results: safety (S). Across all 345 (configuration, operator, α) combinations the bounds hold with the two-standard-error allowance: Theorem 4.2(i)–(ii), in the κ -form and in the closed form, yes; the harmful-event frequency bound, yes; the cap of Corollary 4.3, yes. The largest ratios of the Monte Carlo excess are 0.94 to the κ -bound, 0.40 to its closed form $\varphi(z)\mathbb{E}[s]$, 0.67 to the cap and 0.40 to the regret bound of Theorem 4.2(iv). At $m = 12$ and $\alpha = 0.1$ the cap is 0.0110 (0.11 times the risk of R , 13.3 times smaller than its closed form), and the exact worst case over operators is 0.0100 (Proposition 4.5(c)). Where $\Delta > 0$ almost surely, the comparison with $\mathbb{E}[\Delta^+\pi]$ checks identity (i), not a bound. The identity was checked replicate by replicate: paired z -test in 274 combinations (largest $|z|$ 2.83), Poisson 99% interval for the acceptance count in the 71 with rare acceptance (1 failure, full pooling, $\alpha = 0.05$, $m = 6$, departure 8τ , not reproduced with 50 fresh seeds; details in `results/tables.md`). The frequency of harmful events (the correction used and harmful) of the EB operator never exceeds 25.30% at $\alpha = 0.5$, 3.36% at $\alpha = 0.1$ and 0.25% at $\alpha = 0.01$, against up to 100.0% for always correcting on the split sample; the frequency of using C given that it is harmful never exceeds α (at most 8.42% for $\alpha = 0.1$). Reversion also caps the two bad operators: full pooling has unreverted risk 1.7324 at $\text{snr} = 16$ against 0.1064 after reversion (the per-configuration tables for all operators, with every bound, are in `results/tables.md`, T1, T2 and T4).

Reverting estimators with the guarantee of Theorem 4.2 (EB operator, $m = 12$)

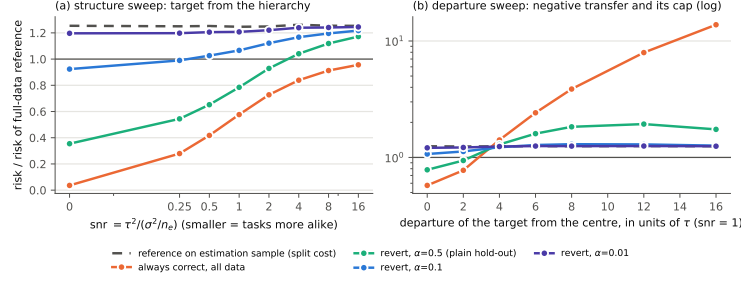


Figure 1: Reverting estimators with the guarantee, EB operator, risks relative to the full-data reference. (a) Structure sweep (Table 3): the reverting estimators start from the split cost (dashed) and recover part of the available gain. (b) Departure sweep (Table 4), log scale: always correcting is harmful from $\kappa = 4$ on; at $\alpha \leq 0.1$ the reverting estimators stay within 1.07–1.30 times the full-data reference, and plain hold-out ($\alpha = 0.5$) reaches 1.93 at $\kappa = 12$. The bound of Theorem 4.2(ii) grows with $\mathbb{E}\|C - R\|$, linearly in κ for the shrinkage operator; what is proved for every κ is the cap of Corollary 4.3, which allows up to 1.38 times the full-data reference at $\alpha = 0.1$ and 2.67 at $\alpha = 0.5$.

Results: usefulness (U). Always correcting with all the data is useful at $\text{snr} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ and, in the hierarchy, never harmful in the criterion’s sense; its relative gain over $\bar{X}_n$ is +42.3% at $\text{snr} = 1$. The reverting estimator *with* the guarantee is useful at $\text{snr} \in \{0, 0.25, 0.5, 1, 2\}$ for $\alpha = 0.5$, $\{0, 0.25, 0.5\}$ for $\alpha = 0.2$, $\{0\}$ for $\alpha = 0.1$, and nowhere for $\alpha \leq 0.05$: at $\text{snr} = 1$ its gain over $\bar{X}_n$ is +21.5% ($\alpha = 0.5$), -6.6% ($\alpha = 0.1$) and -20.8% ($\alpha = 0.01$), while its gain over the reference on the same estimation sample is +37.1%, +14.5% and +3.1%. The difference is the split cost of Corollary 4.4, which the guarantee does not cover, plus the gain forfeited by the test (Remark 4.6): at $\text{snr} = 1$ and $\alpha = 0.1$ the correction is used in 19.6% of the replicates although it helps in 82.3% of them. The held-out-size sweep (`results/tables.md`, T3; τ^2 fixed, so that the estimators that do not use the split stay flat in m) shows the trade-off: with $m = 3$ (a 5% split) the gain of the guaranteed estimator over $\bar{X}_n$ is +3.1% at $\kappa = 0$ and -17.9% at $\kappa = 8$; with $m = 24$ it is -25.0% and -68.8%. The two variants without guarantee are useful over a wider range: refit at $\text{snr} \in \{0, 0.25, 0.5, 1\}$ ($\alpha = 0.1$) and SURE reversion at $\text{snr} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$, the latter within a few points of always correcting in the hierarchy. 4 of the marks in Table 3 are borderline (lower end of the interval within 2 points of the threshold; marked “u”); in particular “refit useful up to $\text{snr} = 1$ ” should not be read as robust (the reduced run, `-fast` with 4000 replicates, keeps every other mark and drops that one).

Results: negative transfer. On the departure sweep always correcting with all the data becomes harmful (upper end of the 95% interval of the gain below zero) at $\kappa \in \{4, 6, 8, 12, 16\}$ and reaches 13.79 times the full-data reference risk at $\kappa = 16$. The guaranteed reverting estimator at $\alpha = 0.1$ stays between 1.07 and 1.30 times the full-data reference over the whole sweep; its excess over R is +0.0042 at $\kappa = 8$ against the bounds 0.0044 (identity), 0.0067 (κ) and 0.0296 (α), and +0.0012 at $\kappa = 16$ against 0.0129 (κ) and the cap 0.0110. The unguaranteed variants are also capped in this design (SURE reversion is harmful only at $\kappa \in \{4, 6, 8\}$, by at most 1.13 times the reference), but nothing proved here says they must be.

Results: transported refit and cross-fitting. In all 345 (configuration, operator, α) combinations the excess of the transported refit over $\bar{X}_n$ stays below the bounds of Theorem 5.1(ii) and (iii) (largest ratios 0.93 and 0.71; 0.87 and 0.66 to the closed forms). Identity (i), checked replicate by replicate, gives $|z| \leq 2.90$ in 274 combinations (0.74 values above 3 expected if it holds exactly); in the 71 rare-acceptance combinations it has 1 Poisson failure, which is the acceptance-count test of Section 7 on the same replicates and in the same configuration, and does not test the term $\eta s \varphi(z + u)$. On an adversarial grid of 560 pairs (θ, C) all bounds hold within two standard errors, and identity (i) gives $|z| \leq 3.09$ over 440 points (1 above the threshold 3 fixed in advance; 1.2 expected if it holds exactly). Without held-out noise (exact quadrature for the families that depend on R only through ξ), the family built from the maximiser of M_η attains 0.9996–1.0000 of the exact cap, the construction of Proposition 4.5(b) 0.34–0.993 and the reflection of (iv) its closed-form value; the other families (fixed shifts, shrinkage), Rao–Blackwellised over the held-out sample, reach at most 0.70 of the cap. The cross-fitted estimator ($m \in \{3, 6, 12\}$) satisfies Corollary 5.2 in all 285 design and 420 adversarial combinations, loosely (largest ratio 0.16 to the bound in the design). It is not free of the split cost either: on the adversarial grid its excess reaches 9.4 times the split cost (fixed additive correction $\|v\| = 2\sigma/\sqrt{m}$, $m = 3$, $\alpha = 0.5$; 0.50 of the risk of $\bar{X}_n$), and 0.07–0.25 times at $\alpha = 0.1$.

In the design (EB operator; per-configuration values in `theory/check_refit_output.txt`) the transported refit pays much less than the split estimator at $m = 12$, but it is not harmless. In the hierarchy its gain over $\bar{X}_n$ at $\alpha = 0.1$ is +6.8% at $\text{snr} = 0$ and +3.4% at $\text{snr} = 1$, and criterion U (applied after the fact to this estimator, added in v0.4) holds at $\text{snr} \in \{0, 0.25, 0.5, 1, 2, 4\}$ for $\alpha = 0.5$, $\{0, 0.25, 0.5, 1\}$ for 0.2 and $\{0\}$ for 0.1. In the held-out-size sweep ($\text{snr} = 1$, $\kappa = 0$, $\alpha = 0.1$) its gain is +4.7% at $m = 3$, but at $m = 24$ it is harmful even with the target inside the hierarchy: excess $+0.0035 \pm 0.0002$ over $\bar{X}_n$ (gain -4.2%), the behaviour that Remark 5.3(e) shows for the oracle. On the departure sweep it is harmful relative to $\bar{X}_n$ at $\kappa \in \{2, 4, 6, 8, 12, 16\}$ for $\alpha = 0.1$,

with risk at most 1.12 times that of $\bar{X}_n$ ($\kappa = 8$; 1.95 at $\alpha = 0.5$, $\kappa = 12$), within the proved bounds. Cross-fitting with $K = 5$ folds gains +29.4% over $\bar{X}_n$ at $\text{snr} = 0$, and its excess on the departure sweep never exceeds +0.0011. The simulation’s refit (operator re-applied) also stays below the bounds of Theorem 5.1 (largest ratios 0.85 and 0.66), but this is empirical only.

Verdict under the predefined criterion. Safety holds everywhere, as it must. Useful transfer with the guaranteed construction occurs only when the tasks are close ($\text{snr} \leq 2$ with plain hold-out, only $\text{snr} = 0$ at $\alpha = 0.1$, never at $\alpha \leq 0.05$), because elsewhere the split costs more than the transfer gains. The transported refit of Section 5 has a bound without the additive split cost and is useful over a wider range with plain hold-out, but it is not better than the split for every correction: with the oracle correction the split beats it for $\alpha = 0.5$ (every m), $\alpha = 0.2$ (every m), $\alpha = 0.1$ (every m) and it is harmful for $\alpha = 0.2$ ($m \in \{24\}$), $\alpha = 0.1$ ($m \in \{24\}$), $\alpha = 0.05$ ($m \in \{24\}$), $\alpha = 0.01$ ($m \in \{12, 24\}$) (Table 1); with the EB operator it is harmful at $m = 24$ even inside the hierarchy; and its worst case retains at least the fraction 4α of the split cost. Cross-fitting reaches 9.4 times the split cost on the adversarial grid. Both bounds are proved here and were checked by an independent internal referee. The other constructions useful over a wide range are classical (empirical Bayes shrinkage with SURE selection; cross-validated selection and refit with the operator re-applied), with no finite-sample “do no harm” bound proved here.

8 Claims and their status

claim	status
No estimator using source data whose law is free of θ beats the minimax risk $d\sigma_\varepsilon^2$ (a); a fixed additive correction is worse by exactly $\mathbb{E}\ b\ ^2$ at every θ (b); for $d \leq 2$ any rule that ever corrects is worse than R somewhere (c); for $d \geq 3$ pointwise but not worst-case improvement exists (d) (Thm. 3.1)	(a) classical, proof included; (b) proved here; (c) literature + proved here; (d) literature
Bayes and frequentist risk of affine shrinkage; available gain $1/(1 + \text{snr})$ (Rem. 3.2)	classical
Held-out loss difference conditionally unbiased, Gaussian with sd $2\sigma\ C - R\ /\sqrt{m}$ (Lem. 4.1); excess-risk identity $\mathbb{E}[\Delta\pi]$ (Thm. 4.2(i))	proved here; identity checked in 345 combinations
Harm $\leq \min\{\alpha\mathbb{E}\Delta^+, \kappa(\alpha)2\sigma\mathbb{E}\ C - R\ /\sqrt{m}\}$, closed form $\varphi(z) \geq \kappa(\alpha)$; $\mathbb{P}(C \text{ used}, \Delta > 0) \leq \alpha\mathbb{P}(\Delta > 0)$; Hoeffding version; near-oracle selection within $(z + \varphi(0))2\sigma\mathbb{E}\ C - R\ /\sqrt{m}$ (Thm. 4.2(ii)–(iv))	proved here; every bound holds everywhere (S)
Cap uniform in θ with constants (κ, κ_2) , for any measurable C (Cor. 4.3)	proved here; holds everywhere
Total cost relative to the full-data reference = split cost (m/n_e of its risk) + controlled term (Cor. 4.4)	proved here
Within one-sided checks at a fixed level α : $\kappa(\alpha)$ is the best constant; for a fixed estimation sample the worst-case harm over operators is of exact order $m^{-1/2}$, with exact value $(\sigma^2/m)\mathbb{E}[M_0(\alpha; \xi)]$ (Prop. 4.5)	proved here; constants and $\mathbb{E}[M_0]$ numerical; (b) checked by Monte Carlo
Reverting estimator = penalised hold-out selection between two candidates; hence no differentiation is to be expected from the safety mechanism itself (Prop. 6.1, Sec. 6)	proved here
Differentiation through the operator family	not explored (three affine instances)
Useful transfer (criterion U) only at $\text{snr} \in \{0, 0.25, 0.5, 1, 2\}$ ($\alpha = 0.5$), $\{0, 0.25, 0.5\}$ (0.2), $\{0\}$ (0.1), never for $\alpha \leq 0.05$, in this design	verified, design-specific; 4 marks borderline
Transported refit: exact identity; bound proportional to $\ C - R\ $, without split-cost term; cap uniform in θ and C , never below 4α times the split cost (Thm. 5.1); cross-fitted version (Cor. 5.2)	proved here; independent internal referee (round 3); constants numerical; bounds hold (MC)
Not better than the split for every correction: with the oracle $C = \theta$ the split is better for $\alpha = 0.5$ (every m), $\alpha = 0.2$ (every m), $\alpha = 0.1$ (every m), and the refit is harmful for $\alpha = 0.2$ ($m \in \{24\}$), $\alpha = 0.1$ ($m \in \{24\}$), $\alpha = 0.05$ ($m \in \{24\}$), $\alpha = 0.01$ ($m \in \{12, 24\}$); with the EB operator harmful at $m = 24$, $\kappa = 0$ (Rem. 5.3(e), Sec. 7)	oracle: exact (proved formula, quadrature); EB: empirical
Worst case of the transported refit below that of the split estimator for $\alpha = 0.1$ ($m \in \{6, 12, 24\}$), $\alpha = 0.05$ (every m), $\alpha = 0.01$ (every m), above for $\alpha = 0.5$ (every m), $\alpha = 0.2$ (every m), $\alpha = 0.1$ ($m \in \{3\}$) (Table 1); at $\alpha = 0.5$ also above in 36 of 36 other designs	numerical (quadrature); not proved in general
Cross-fitting is not free of the split cost: up to 9.4 times it on the adversarial grid	empirical (MC)
“ C if the check passes, else $\bar{X}_n$ ” keeps 0.32–1.99 of the split cost (Rem. 5.3(c))	derived here; quadrature
Refit as simulated (operator re-applied to $\bar{X}_n$) and SURE reversion useful over a wider range and capped in this design	empirical; no guarantee (not covered by Thm. 5.1)
General accuracy improvement, transfer guarantee, statements about real populations	not claimed

Table 5: Claims and their status. “Verified” and “holds” mean reproduced by the script of Appendix A with the fixed seed; neither is a proof.

Limitations. The model is Gaussian location with known variance; the κ -bound and the cap use Gaussian held-out observations, the Hoeffding version needs bounded ones, and the t -test variant is only remarked on. The reference is the sample mean; for a biased reference (a shared instrument bias, say) the held-out sample of the target carries the same bias and the check cannot see the correction, so that setting needs unbiased calibration data and is not covered. The simulation has one design ($d = 5$, $n = 60$, $S = 30$) and one structural family; the “useful” region depends on the split fraction and on the dimension through the ratio of the κ -bound

to the reference risk, of order $\sqrt{n_e/m}/\sqrt{d}$ for the shrinkage operator, and would move in another design. The unguaranteed variants are reported, not analysed. Theorem 5.1 and Corollary 5.2 were derived in a separate pass, re-derived by the integrating pass, checked by Monte Carlo and checked by an independent internal referee (round 3); for the transported refit no analogue of the α -form or of the harmful-event bound is proved. The transported refit adds to $\bar{X}_n$ a vector calibrated for R : when $m \geq n_e$ ($\eta \geq \frac{1}{2}$) even the oracle correction is never better than $\bar{X}_n$ (accepted always, its risk is $d\sigma^2 m/(nn_e)$ against $d\sigma^2/n$); the design ($m/n_e = 12/48$) is on the favourable side. Nothing is said about real data.

Next steps for the research line. (a) *Partially closed in v0.4* (Section 5): the transported refit and its cross-fitted version have finite-sample bounds; open are a bound for the refit as simulated (operator re-applied), an α -form and a harmful-event bound for the transported refit, a cross-fitting bound that exploits the averaging instead of Jensen’s inequality, and conditions on the correction under which the transported refit beats the split. (b) A selection bound for SURE reversion with the affine operator, whose selection region is a ball around $\hat{\mu}$ and whose risk is not given by SURE because of the discontinuity on the sphere. (c) If the original operator definitions of OMR001 are recovered, place each in Table 2 and compute $\mathbb{E}\|C - R\|$ for it; by Theorem 4.2 that is all the safety analysis needs, and such a family is the only remaining source of differentiation. (d) Decide whether the line should be closed with this note or continued only along (a)–(b).

A Reproducibility

The folder contains `experiments/safe_reversion.py` (all sweeps, constants, identity re-checks, tables and figures; `-fast` runs a reduced version) and `experiments/make_numbers.py`, which converts `results/results.json` into the macros and table bodies of this document, so that every number in the text is generated and none is typed. The full run used seed 20260930 (re-checks: 20260930 + 1 and 20260930 + 1000 + i), Python 3.11.15, NumPy 2.4.6 [8], SciPy 1.17.1 [19] and Matplotlib 3.11.2 [10], and took 14 s of wall-clock time (14 s of CPU) on a shared four-core machine. `experiments/check_uniform_cap.py` (seconds; no number in the text comes from it) checks Corollary 4.3 on a grid and Proposition 4.5(b) by Monte Carlo. Since v0.4, `theory/check_refit.py` (20 000 replicates per point, seeds 20260930, + 7, + 11, + 13; 86 s of CPU) checks Section 5 and writes `results/refit_check.json`, whose SHA-256 (prefix 944fc82f9ad3a6fe) is recorded in `results/refit_check.json.sha256` and verified by `make_numbers.py`. Since v0.5 it also computes by quadrature, without new draws, the exact quantities of Proposition 4.5(c), Table 1 and Remark 5.3, and M_η through the collinear reduction of Theorem 5.1(iv) (relative change of the exact cap at most 7×10^{-7}).

Procedure. Criterion U was fixed in writing before the runs (script `docstring` and `results.json`). The two-standard-error tolerance of S and the thresholds of the identity check ($|z| \leq 3$, Poisson 99%) were chosen before the final run but not recorded beforehand. The Poisson comparison replaced a paired z -test after a first run showed that the normal approximation fails where acceptance is rare. After the first internal review the held-out-size sweep was re-run with τ^2 fixed and the borderline mark was introduced; after the second, checks against the κ -form of Theorem 4.2(ii) and against the cap with (κ, κ_2) were added, and maxima quoted as “never exceeds” are rounded up; after the third, exact quadrature (or Rao–Blackwellised estimates) replaced the bracket of Table 1 and the raw Monte Carlo ratios of the adversarial families, and maxima of the v0.4 quantities are also rounded up; no simulated number changed. No other change to the protocol or the criteria was made after seeing results. Proposition 4.5(b) and the constants κ, κ_2 in Theorem 4.2(ii) and Corollary 4.3 are due to the second internal review; Proposition 4.5(c), the oracle formula of Remark 5.3(e) and the collinear form of Theorem 5.1(iv) to the third. Risks are Monte Carlo means with standard errors from the replicate standard deviation; relative gains use paired differences and treat the denominator as fixed (its relative standard error is below 1%); $\kappa(\alpha)$ and $\kappa_2(\alpha)$ are computed by `scipy.optimize.minimize_scalar` on $[0, 20]$, which contains both maximisers (at most 0.75 and 1.19), with tolerance 10^{-10} . The draws per replicate are listed in the README; the constants $\kappa, u^*, \kappa_2, \varphi(z_{1-\alpha})/\kappa$ and $z_{1-\alpha} + \varphi(0)$ for every α in `results/tables.md` (T5; $\varphi(1) = 0.2420$).

B Proofs for Section 5

Proof of Theorem 5.1. Write $h = \bar{Y} - \theta$, so that $\bar{X}_n - \theta = (1 - \eta)e + \eta h$ and, because $J^2 = J$ and $\Delta = \|w\|^2 + 2\langle w, e \rangle$,

$$\ell(\hat{\theta}_{\text{rf}}) - \ell(\bar{X}_n) = J(\|w\|^2 + 2\langle w, \bar{X}_n - \theta \rangle) = J((1 - \eta)\Delta + \eta\|w\|^2 + 2\eta\langle w, h \rangle).$$

This difference is at least $-\ell(\bar{X}_n)$, so all expectations below are well defined; (iii) shows that they are finite.

(i) By Lemma 4.1, given $(\mathcal{F}, \theta)$ we have $\langle w, h \rangle = (s/2)Z$ with $Z \sim N(0, 1)$ and $\hat{D} = \Delta - sZ$, so $J = \mathbf{1}\{Z \geq z + u\}$: the check depends on the held-out sample only through Z . Hence $\mathbb{E}[J \mid \mathcal{F}, \theta] = \pi$ and, since $\int_{\tau}^{\infty} x\varphi(x) dx = \varphi(\tau)$, $\mathbb{E}[J\langle w, h \rangle \mid \mathcal{F}, \theta] = (s/2)\varphi(z + u)$. Take expectations. The second form uses $(1 - \eta)\Delta + \eta\|w\|^2 = \Delta - 2\eta\langle w, e \rangle$. If $s = 0$, then $w = 0$ and both sides vanish.

(ii) Since $\|w\|^2 - \Delta = -2\langle w, e \rangle \leq 2\|w\|\|e\| = s\xi$, we have $(1 - \eta)\Delta + \eta\|w\|^2 \leq s(u + \eta\xi)$, so the conditional excess is at most $s g(u)$ with $g(u) = (u + x)\Phi(-(z + u)) + \eta\varphi(z + u)$ and $x = \eta\xi$; this gives the first inequality.

The second follows from $\Psi_\eta(\alpha; x) \leq \Psi_\eta(\alpha; 0) + x$ and $s\xi = 2\|w\|\|e\|$. For the closed form: if $u \geq 0$, then $g(u) \leq \kappa(\alpha) + \eta\varphi(z) + x\alpha$ (because $z \geq 0$, φ decreases on $[0, \infty)$ and $\Phi(-(z+u)) \leq \alpha$); if $-x \leq u < 0$, then $g(u) \leq \eta\varphi(0) + x\Phi(x-z)$; if $u < -x$, the first term of g is negative and $g(u) \leq \eta\varphi(0)$. In all three cases $g(u) \leq \bar{\kappa}_\eta + x\Phi(x-z)$, because $\alpha \leq \Phi(x-z)$.

(iii) Put $\|w\| = \sigma\omega/\sqrt{m}$ and $\Delta = \sigma^2\delta/m$. Then $s = 2\sigma^2\omega/m$, $u = \delta/(2\omega)$, the conditional excess in (i) equals $(\sigma^2/m)f_\eta(\omega, \delta)$, and $|\Delta - \|w\|^2| = 2|\langle w, e \rangle| \leq 2\|w\|\|e\|$ reads $|\delta - \omega^2| \leq 2\xi\omega$. Since ξ is a function of $e \sim N(0, \sigma^2 I_d/n_e)$, whose law does not depend on θ , this proves the first claim. For the closed form, $f_\eta \leq 2\omega g(u)$ with $x = \eta\xi$, as in (ii). If $\omega \leq 2\xi$, then $f_\eta \leq 4\xi g(u)^+ \leq 4\xi(\bar{\kappa}_\eta + \eta\xi\Phi(\eta\xi - z))$. If $\omega > 2\xi$, then $u \geq \omega/2 - \xi > 0$, so $2\omega \leq 4\xi + 4u$ and $g(u) \geq 0$; hence $f_\eta \leq 4\xi g(u) + 4u g(u) \leq 4\xi(\bar{\kappa}_\eta + \eta\xi\alpha) + 4(\kappa_2 + \eta\xi\kappa + \eta\kappa_\varphi)$, using $u^2\Phi(-(z+u)) \leq \kappa_2$, $u\Phi(-(z+u)) \leq \kappa$ and $u\varphi(z+u) \leq \kappa_\varphi$, with $\kappa_\varphi \leq \sup_{t \geq 0} t\varphi(t) = \varphi(1)$ since $z \geq 0$. Both case bounds are at most the stated bound on M_η . Finally, with $R = \bar{X}_e$, $\xi \sim \sqrt{m/n_e}\|Z\|$, so $\mathbb{E}\xi = c_d\sqrt{m/n_e}$, $\mathbb{E}\xi^2 = dm/n_e$ and $\mathbb{E}\xi^3 = \mu_{3,d}(m/n_e)^{3/2}$; also $\Phi(\eta\xi - z) \leq \alpha + \eta\xi\varphi(0)$. Multiply by σ^2/m ; the term $(\sigma^2/m)4\eta\alpha dm/n_e$ equals $4\alpha d\sigma^2 m/(nn_e)$.

(iv) Fix $\xi > 0$ and put $u = \delta/(2\omega)$. For fixed u , $f_\eta = \eta\Phi(-(z+u))\omega^2 + 2\omega[(1-\eta)u\Phi(-(z+u)) + \eta\varphi(z+u)]$ is a convex quadratic in ω , and the constraint $|\delta - \omega^2| \leq 2\xi\omega$ reads $2(u-\xi) \leq \omega \leq 2(u+\xi)$ with $\omega > 0$. The supremum over this interval is therefore attained at an end point: $\omega = 2(u+\xi)$, i.e. $\delta = \omega^2 - 2\xi\omega$; $\omega = 2(u-\xi)$ when $u > \xi$, i.e. $\delta = \omega^2 + 2\xi\omega$; or the limit $\omega \rightarrow 0$, where $f_\eta \rightarrow 0$. Hence $M_\eta(\alpha; \xi) = \max\{0, \sup_{\omega > 0, c = \pm 1} F_c(\omega, \xi)\}$ with

$$F_c(\omega, \xi) = f_\eta(\omega, \omega^2 + 2c\xi\omega) = (\omega^2 + 2(1-\eta)c\xi\omega)\Phi(-(z + \frac{\omega}{2} + c\xi)) + 2\eta\omega\varphi(z + \frac{\omega}{2} + c\xi),$$

which is jointly continuous, tends to 0 as $\omega \rightarrow 0$ and, uniformly for ξ in compacts, as $\omega \rightarrow \infty$; so $M_\eta(\alpha; \cdot) \geq 0$ is continuous. Fix $\theta = \theta_0$ and $\varepsilon > 0$. Choose Ξ with $(\sigma^2/m)\mathbb{E}[M_\eta(\alpha; \xi)\mathbf{1}\{\xi > \Xi\}] < \varepsilon/2$ (possible by the closed form of (iii)), put $w = 0$ where $\xi > \Xi$, and on a fine enough finite partition of $[0, \Xi]$ choose (ω, c) with $c \in \{-1, 1\}$ constant on each cell (or $w = 0$ on the cell) so that the conditional excess is at least $(\sigma^2/m)M_\eta - \varepsilon/2$ on it (uniform continuity on a compact). With $\hat{e} = e/\|e\|$, the vector $w = (\sigma/\sqrt{m})\omega c\hat{e} = (c\omega/\xi)(R - \theta_0)$ realises (ω, δ) in every dimension $d \geq 1$, and $C = R + w$ is a measurable function of R , hence $\mathcal{F}$ -measurable: the worst corrections inflate ($c = 1$) or shrink and overshoot ($c = -1$) the error of R . For the explicit pair, $C = 2\theta_0 - R$ gives $w = -2e$, $\Delta = 0$, $\pi = \alpha$, $(1-\eta)\Delta + \eta\|w\|^2 = 4\eta\|e\|^2$ and $s = 4\sigma\|e\|/\sqrt{m}$; apply (i) with $\mathbb{E}\|e\|^2 = d\sigma^2/n_e$ and $\mathbb{E}\|e\| = \sigma c_d/\sqrt{n_e}$. The second term of the result is positive, so the supremum over C is at least 4α times the split cost. $\square$

Proof of Corollary 5.2. Each observation lies outside exactly $K - 1$ folds, so $\frac{1}{K} \sum_k R_k = \bar{X}_n$. By convexity of the squared loss, $\ell(\hat{\theta}_{\text{cf}}) \leq \frac{1}{K} \sum_k \ell(\bar{X}_n + J_k(C_k - R_k))$. The k -th term is the transported refit of Theorem 5.1 with estimation sample outside I_k , held-out sample I_k , $\eta = m/n = 1/K$ and the same law of ξ . $\square$

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